

Product Batch - Task Submission - 02.09.2026

Task - 1

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1. Delete a node from a binary search tree.

```
main.py Visualize Run ▶
211 # =====
212 # QUESTION 1
213 # =====
214
215 print("\n1. DELETE A NODE FROM BST")
216
217 delete_value = 70
218
219 print("Before deleting", delete_value, ":", inorder(root))
220
221 root = delete_node(root, delete_value)
222
223 print("After deleting", delete_value, ":", inorder(root))
224
```

2. Check if a binary tree is a binary search tree.

```
main.py Visualize Run ▶
225
226 # =====
227 # QUESTION 2
228 # =====
229
230 print("\n2. CHECK IF BINARY TREE IS BST")
231
232 if is_bst(root):
233     print("The given tree is a BST")
234 else:
235     print("The given tree is NOT a BST")
236
237
238 # =====
239 # QUESTION 3
```

3. Find the kth smallest and kth largest elements in a BST.

```
main.py Visualize Run ▶
237
238 # =====
239 # QUESTION 3
240 # =====
241
242 print("\n3. KTH SMALLEST AND KTH LARGEST")
243
244 k = 3
245
246 smallest = kth_smallest(root, k)
247 largest = kth_largest(root, k)
248
249 print("Inorder:", inorder(root))
250
251 print(k, "th smallest element:", smallest)
252 print(k, "th largest element :", largest)
253
```

4. Find whether the given BST is balanced or not.

```
main.py Visualize Run ▶
254
255 # =====
256 # QUESTION 4
257 # =====
258
259 print("\n4. CHECK WHETHER BST IS BALANCED")
260
261 if is_balanced(root):
262     print("The BST is Balanced")
263 else:
264     print("The BST is NOT Balanced")
265 |
266 
267
```

OUTPUT

```
=====
BINARY SEARCH TREE
=====
```

Original BST Inorder:

[10, 20, 25, 30, 40, 50, 60, 70, 80, 90]

1. DELETE A NODE FROM BST

Before deleting 70 : [10, 20, 25, 30, 40, 50, 60, 70, 80, 90]

After deleting 70 : [10, 20, 25, 30, 40, 50, 60, 80, 90]

2. CHECK IF BINARY TREE IS BST

The given tree is a BST

3. KTH SMALLEST AND KTH LARGEST

Inorder: [10, 20, 25, 30, 40, 50, 60, 80, 90]

3 th smallest element: 25

3 th largest element : 60

4. CHECK WHETHER BST IS BALANCED

The BST is Balanced

=== Code Execution Successful ===